

Random Effects

Random Effects Models – Exam Questions with Solutions

Question 1: Fixed vs Random Effects

Explain the difference between fixed and random effects. Give an example where a factor should be treated as random.

Solution:

- **Fixed effects:** Levels are specifically chosen and of direct interest. Inference is about these levels.
- **Random effects:** Levels are sampled from a larger population. Inference is about population variability.

Example: Machines in a factory.

- Fixed: if only those machines matter.
- Random: if machines are sampled from a large population and we want to generalize.

Question 2: Model Specification

Consider a study with a treatments and b groups (random). Write the random effects model and define all terms.

Solution:

$$Y_{ij} = \mu + \tau_i + b_j + \epsilon_{ij}$$

- μ : overall mean
- τ_i : fixed effect of treatment i
- $b_j \sim N(0, \sigma_b^2)$: random group effect
- $\epsilon_{ij} \sim N(0, \sigma^2)$: error term

Question 3: Mean and Variance

Given the model $Y_{ij} = \mu + \tau_i + b_j + \epsilon_{ij}$, compute $E(Y_{ij})$ and $\text{Var}(Y_{ij})$.

Solution:

$$E(Y_{ij}) = \mu + \tau_i$$

$$\text{Var}(Y_{ij}) = \text{Var}(b_j) + \text{Var}(\epsilon_{ij}) = \sigma_b^2 + \sigma^2$$

Question 4: Covariance Structure

Compute $\text{Cov}(Y_{ij}, Y_{i'j'})$ for:

1. Same group ($j = j'$)
2. Different groups ($j \neq j'$)

Solution:

- Same group:
$$\text{Cov}(Y_{ij}, Y_{ij'}) = \sigma_b^2$$
- Different groups:
$$\text{Cov}(Y_{ij}, Y_{i'j'}) = 0$$

Question 5: Intraclass Correlation (ICC)

Define and interpret the intraclass correlation coefficient.

Solution:

$$\rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}$$

Interpretation:

- $\rho \approx 0$: little within-group similarity
- ρ large: strong within-group correlation

Question 6: When to Use Random Effects

List three reasons why a random effects model is preferred over a fixed effects model.

Solution:

- Interest is in population variability, not specific levels
- Want to generalize to new/unseen groups
- Need to model correlation within groups

Question 7: Interpretation of Variance Components

Explain the meaning of σ_b^2 and σ^2 .

Solution:

- σ_b^2 : between-group variability
- σ^2 : within-group (residual) variability

Question 8: Matrix Form

Write the random effects model in matrix form and give the covariance of Y .

Solution:

$$Y = X\beta + Zb + \epsilon$$

$$b \sim N(0, \sigma_b^2 I), \quad \epsilon \sim N(0, \sigma^2 I)$$

$$\text{Var}(Y) = Z\sigma_b^2 Z^T + \sigma^2 I$$

Question 9: Interpretation Question

Why does the random effect not affect the mean but affect the covariance?

Solution:

Because:

$$E(b_j) = 0$$

So it cancels out in expectation:

$$E(Y_{ij}) = \mu + \tau_i$$

However, b_j contributes to variability:

$$\text{Var}(Y_{ij}) = \sigma_b^2 + \sigma^2$$

Thus, it affects covariance but not mean.

Question 10: Conceptual

What would happen if we incorrectly treat a random effect as fixed?

Solution:

- Inference becomes conditional on observed levels only
- Cannot estimate variance components
- Cannot generalize to new levels

Question 11: Correlation Interpretation

If $\rho = 0.8$, what does this imply?

Solution:

Strong within-group similarity. Most variability is explained by group differences.

Question 12: Estimation

Why is OLS not appropriate for random effects models?

Solution:

Because observations are correlated:

$$\text{Cov}(Y_i, Y_j) \neq 0$$

OLS assumes independence. Instead, use GLS:

$$\hat{\beta} = (X^T V^{-1} X)^{-1} X^T V^{-1} Y$$

where $V = \text{Var}(Y)$.

Question 13: Design Question

Give an example of a nested random effects structure.

Solution:

Students nested within classrooms:

$$Y_{ijk} = \mu + \tau_i + b_j + \epsilon_{ijk}$$

where:

- b_j : classroom random effect
- students within same classroom are correlated

Random Effects ANOVA and Degrees of Freedom

Question 14: One-Way Random Effects Model

Suppose there are a randomly selected groups, with n observations per group.

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

where

$$\alpha_i \sim N(0, \sigma_\alpha^2), \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

Find the ANOVA table.

Solution:

Total number of observations:

$$N = an$$

Degrees of freedom:

$$df_{\text{Group}} = a - 1$$

$$df_E = a(n - 1)$$

$$df_T = an - 1$$

Sum of squares:

$$SS_{\text{Group}} = n \sum_{i=1}^a (\bar{Y}_i - \bar{Y}_{..})^2$$

$$SS_E = \sum_{i=1}^a \sum_{j=1}^n (Y_{ij} - \bar{Y}_i)^2$$

$$SS_T = \sum_{i=1}^a \sum_{j=1}^n (Y_{ij} - \bar{Y}_{..})^2$$

Mean squares:

$$MS_{\text{Group}} = \frac{SS_{\text{Group}}}{a - 1}$$

$$MS_E = \frac{SS_E}{a(n - 1)}$$

Expected mean squares:

$$E(MS_E) = \sigma^2$$

$$E(MS_{\text{Group}}) = \sigma^2 + n\sigma_\alpha^2$$

ANOVA table:

Source	SS	df	MS	EMS
Group	SS_{Group}	$a - 1$	MS_{Group}	$\sigma^2 + n\sigma_\alpha^2$
Error	SS_E	$a(n - 1)$	MS_E	σ^2
Total	SS_T	$an - 1$		

To test whether the random group effect exists:

$$H_0 : \sigma_\alpha^2 = 0$$

$$F_0 = \frac{MS_{\text{Group}}}{MS_E}$$

with degrees of freedom:

$$F_0 \sim F_{a-1, a(n-1)}$$

Question 15: Estimate Variance Components

Using the one-way random effects model, estimate σ^2 and σ_α^2 .

Solution:

Since

$$E(MS_E) = \sigma^2$$

we estimate

$$\hat{\sigma}^2 = MS_E$$

Since

$$E(MS_{\text{Group}}) = \sigma^2 + n\sigma_\alpha^2$$

we estimate

$$\hat{\sigma}_\alpha^2 = \frac{MS_{\text{Group}} - MS_E}{n}$$

If this value is negative, it is usually set to 0:

$$\hat{\sigma}_\alpha^2 = \max\left\{0, \frac{MS_{\text{Group}} - MS_E}{n}\right\}$$

Question 16: Intraclass Correlation from ANOVA

Using the one-way random effects model, estimate the intraclass correlation coefficient.

Solution:

The ICC is

$$\rho = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2}$$

Using ANOVA estimates:

$$\hat{\rho} = \frac{\hat{\sigma}_\alpha^2}{\hat{\sigma}_\alpha^2 + \hat{\sigma}^2}$$

where

$$\hat{\sigma}^2 = MS_E$$

and

$$\hat{\sigma}_\alpha^2 = \frac{MS_{\text{Group}} - MS_E}{n}$$

Therefore,

$$\hat{\rho} = \frac{MS_{\text{Group}} - MS_E}{MS_{\text{Group}} + (n-1)MS_E}$$

Question 17: Two-Way Model with One Fixed Factor and One Random Block Factor

Suppose treatment is fixed and block is random:

$$Y_{ijk} = \mu + \tau_i + b_j + \epsilon_{ijk}$$

where

$$i = 1, \dots, a, \quad j = 1, \dots, b, \quad k = 1, \dots, n$$

and

$$b_j \sim N(0, \sigma_b^2), \quad \epsilon_{ijk} \sim N(0, \sigma^2)$$

Find the ANOVA degrees of freedom.

Solution:

Total number of observations:

$$N = abn$$

Treatment degrees of freedom:

$$df_A = a - 1$$

Random block degrees of freedom:

$$df_B = b - 1$$

Error degrees of freedom:

$$df_E = ab(n - 1)$$

Total degrees of freedom:

$$df_T = abn - 1$$

ANOVA table:

Source	df	MS	EMS
Treatment <i>A</i>	$a - 1$	MS_A	$\sigma^2 + bn \sum \frac{\tau_i^2}{a-1}$
Block <i>B</i>	$b - 1$	MS_B	$\sigma^2 + an \sigma_b^2$
Error	$ab(n - 1)$	MS_E	σ^2
Total	$abn - 1$		

Tests:

For treatment effect:

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_a = 0$$

$$F_0 = \frac{MS_A}{MS_E}$$

For random block effect:

$$H_0 : \sigma_b^2 = 0$$

$$F_0 = \frac{MS_B}{MS_E}$$

Question 18: Two-Way Random Effects Model

Suppose both factors are random:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \epsilon_{ijk}$$

where

$$i = 1, \dots, a, \quad j = 1, \dots, b, \quad k = 1, \dots, n$$

and

$$A_i \sim N(0, \sigma_A^2), \quad B_j \sim N(0, \sigma_B^2), \quad (AB)_{ij} \sim N(0, \sigma_{AB}^2),$$

Find the ANOVA table.

Solution:

Total observations:

$$N = abn$$

Degrees of freedom:

$$df_A = a - 1$$

$$df_B = b - 1$$

$$df_{AB} = (a - 1)(b - 1)$$

$$df_E = ab(n - 1)$$

$$df_T = abn - 1$$

Expected mean squares:

$$E(MS_E) = \sigma^2$$

$$E(MS_{AB}) = \sigma^2 + n\sigma_{AB}^2$$

$$E(MS_A) = \sigma^2 + n\sigma_{AB}^2 + bn\sigma_A^2$$

$$E(MS_B) = \sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$$

ANOVA table:

Source	df	MS	EMS
A	$a - 1$	MS_A	$\sigma^2 + n\sigma_{AB}^2 + bn\sigma_A^2$
B	$b - 1$	MS_B	$\sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$
AB	$(a - 1)(b - 1)$	MS_{AB}	$\sigma^2 + n\sigma_{AB}^2$
E	$ab(n - 1)$	MS_E	σ^2
T	$abn - 1$		

Tests:

For factor A :

$$H_0 : \sigma_A^2 = 0$$

$$F_0 = \frac{MS_A}{MS_{AB}}$$

For factor B :

$$H_0 : \sigma_B^2 = 0$$

$$F_0 = \frac{MS_B}{MS_{AB}}$$

For interaction:

$$H_0 : \sigma_{AB}^2 = 0$$

$$F_0 = \frac{MS_{AB}}{MS_E}$$

Question 19: Two-Way Mixed Effects Model

Suppose factor A is fixed and factor B is random:

$$Y_{ijk} = \mu + \tau_i + B_j + (\tau B)_{ij} + \epsilon_{ijk}$$

where

$$i = 1, \dots, a, \quad j = 1, \dots, b, \quad k = 1, \dots, n$$

A is fixed, while B and AB are random.

Find the ANOVA table.

Solution:

Degrees of freedom:

$$df_A = a - 1$$

$$df_B = b - 1$$

$$df_{AB} = (a - 1)(b - 1)$$

$$df_E = ab(n - 1)$$

$$df_T = abn - 1$$

Expected mean squares:

$$E(MS_E) = \sigma^2$$

$$E(MS_{AB}) = \sigma^2 + n\sigma_{AB}^2$$

$$E(MS_B) = \sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$$

$$E(MS_A) = \sigma^2 + n\sigma_{AB}^2 + bn \frac{\sum_{i=1}^a \tau_i^2}{a - 1}$$

ANOVA table:

Source	df	MS	EMS
A	$a - 1$	MS_A	$\sigma^2 + n\sigma_{AB}^2 + bn \frac{\sum \tau_i^2}{a - 1}$
B	$b - 1$	MS_B	$\sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$
AB	$(a - 1)(b - 1)$	MS_{AB}	$\sigma^2 + n\sigma_{AB}^2$
E	$ab(n - 1)$	MS_E	σ^2
T	$abn - 1$		

Tests:

For fixed factor A :

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_a = 0$$

$$F_0 = \frac{MS_A}{MS_{AB}}$$

For random factor B :

$$H_0 : \sigma_B^2 = 0$$

$$F_0 = \frac{MS_B}{MS_{AB}}$$

For interaction:

$$H_0 : \sigma_{AB}^2 = 0$$

$$F_0 = \frac{MS_{AB}}{MS_E}$$

Question 20: Degree of Freedom Practice

A one-way random effects experiment has $a = 5$ groups and $n = 4$ observations per group. Find all degrees of freedom.

Solution:

Total observations:

$$N = an = 5(4) = 20$$

Group degrees of freedom:

$$df_{\text{Group}} = a - 1 = 5 - 1 = 4$$

Error degrees of freedom:

$$df_E = a(n-1) = 5(4-1) = 15$$

Total degrees of freedom:

$$df_T = N - 1 = 20 - 1 = 19$$

Check:

$$4 + 15 = 19$$

Question 21: Degree of Freedom Practice for Two-Way Model

Suppose $a = 3$, $b = 4$, and $n = 2$.

Find the degrees of freedom for A , B , AB , error, and total.

Solution:

Total observations:

$$N = abn = 3(4)(2) = 24$$

$$df_A = a - 1 = 3 - 1 = 2$$

$$df_B = b - 1 = 4 - 1 = 3$$

$$df_{AB} = (a-1)(b-1) = (3-1)(4-1) = 6$$

$$df_E = ab(n-1) = 3(4)(2-1) = 12$$

$$df_T = N - 1 = 24 - 1 = 23$$

Check:

$$2 + 3 + 6 + 12 = 23$$

Question 22: No Replication Case

Suppose a two-way design has one observation per cell, so $n = 1$.

What is the error degree of freedom?

Solution:

Error degrees of freedom:

$$df_E = ab(n-1)$$

Since $n = 1$,

$$df_E = ab(1-1) = 0$$

So there is no pure error degree of freedom.

In this case, the interaction term is often used as the error term if interaction is assumed negligible.

Question 23: Which Mean Square Should Be the Denominator?

In a two-way random effects model with replication, what denominator should be used to test factor A ?

Solution:

For the two-way random effects model:

$$E(MS_A) = \sigma^2 + n\sigma_{AB}^2 + bn\sigma_A^2$$

$$E(MS_{AB}) = \sigma^2 + n\sigma_{AB}^2$$

The only difference is the $bn\sigma_A^2$ term.

Therefore, to test

$$H_0 : \sigma_A^2 = 0$$

use

$$F_0 = \frac{MS_A}{MS_{AB}}$$

Question 24: Variance Component Estimation in Two-Way Random Effects

For the two-way random effects model, estimate the variance components.

Solution:

From the EMS:

$$E(MS_E) = \sigma^2$$

$$E(MS_{AB}) = \sigma^2 + n\sigma_{AB}^2$$

$$E(MS_A) = \sigma^2 + n\sigma_{AB}^2 + bn\sigma_A^2$$

$$E(MS_B) = \sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$$

Therefore:

$$\hat{\sigma}^2 = MS_E$$

$$\hat{\sigma}_{AB}^2 = \frac{MS_{AB} - MS_E}{n}$$

$$\hat{\sigma}_A^2 = \frac{MS_A - MS_{AB}}{bn}$$

$$\hat{\sigma}_B^2 = \frac{MS_B - MS_{AB}}{an}$$

Question 25: Quick Rules for Degrees of Freedom

Solution:

For a two-factor design with replication:

a = number of levels of factor A

b = number of levels of factor B

n = replicates per cell

Then:

$$df_A = a - 1$$

$$df_B = b - 1$$

$$df_{AB} = (a - 1)(b - 1)$$

$$df_E = ab(n - 1)$$

$$df_T = abn - 1$$

Always check:

$$df_A + df_B + df_{AB} + df_E = df_T$$